



SOC HOME LAB WORKSHOP REPORT



Hands-On Security Operations Center Simulation
Threat Detection | Log Analysis | Incident Monitoring

Lab Overview

Lab Type	SOC Home Lab Simulation
SIEM Platform	Wazuh v4.7.5
Victim Machine	Ubuntu (ranime-VMware-Virtual-Platform)
Attacker Machine	Kali Linux 2025.2 (kali-lab, Agent 002)
Network	192.168.23.0/24 (VMware Virtual Network)
Date	February 21, 2026
Objective	Simulate SSH brute-force attack & monitor via SIEM

1. Introduction

This workshop report documents a hands-on SOC (Security Operations Center) home lab built to simulate real-world cyberattacks and practice Tier-1 analyst skills including threat detection, log analysis, and incident monitoring.

The lab environment replicates a simplified enterprise network with a victim machine, an attacker machine, and a centralized SIEM (Security Information and Event Management) platform to collect and analyze security events in real time.

2. Lab Architecture

2.1 Components

- Ubuntu Server — Victim machine (agent: ranime-VMware-Virtual-Platform) running SSH service
- Kali Linux 2025.2 — Attacker machine (agent: kali-lab, ID 002, IP 192.168.23.141)
- Wazuh Manager (v4.7.5) — SIEM deployed at 192.168.23.138, collecting and analyzing logs
- VMware Virtual Platform — Hypervisor hosting all virtual machines

2.2 Network Topology

All machines were deployed on the same VMware virtual network (192.168.23.0/24), enabling direct communication between attacker and victim for the SSH brute-force simulation. The Wazuh manager monitored the victim via a deployed agent.

3. Attack Simulation: SSH Brute Force

3.1 Methodology

An SSH brute-force attack was launched from the Kali Linux machine targeting the Ubuntu victim. The attack involved repeated automated login attempts with different credentials, a common technique catalogued under MITRE ATT&CK.

3.2 MITRE ATT&CK Mapping

- Tactic: Credential Access
- Technique: Password Guessing (T1110.001)
- Technique: Brute Force (T1110)
- Defense Evasion — observed as top tactic on kali-lab agent

3.3 Attack Results

- Total SSH-related alerts generated: 8
- Level 12 or above (critical) alerts: 0
- Authentication failures: 8
- Authentication successes: 0 (attack was unsuccessful)
- Primary Wazuh rule triggered: 5503 — PAM: User login failed
- Secondary rule triggered: 2502 — syslog: User missed the password

The attack produced a visible spike in the Alert Level Evolution chart around 12:00, corresponding to the burst of brute-force attempts.

4. Wazuh SIEM Monitoring

4.1 Security Events Dashboard

The Wazuh Security Events dashboard was filtered using the keyword "ssh" to isolate SSH-related alerts. The dashboard presented real-time statistics including total alert count, severity breakdown, and the Alert Level Evolution chart.

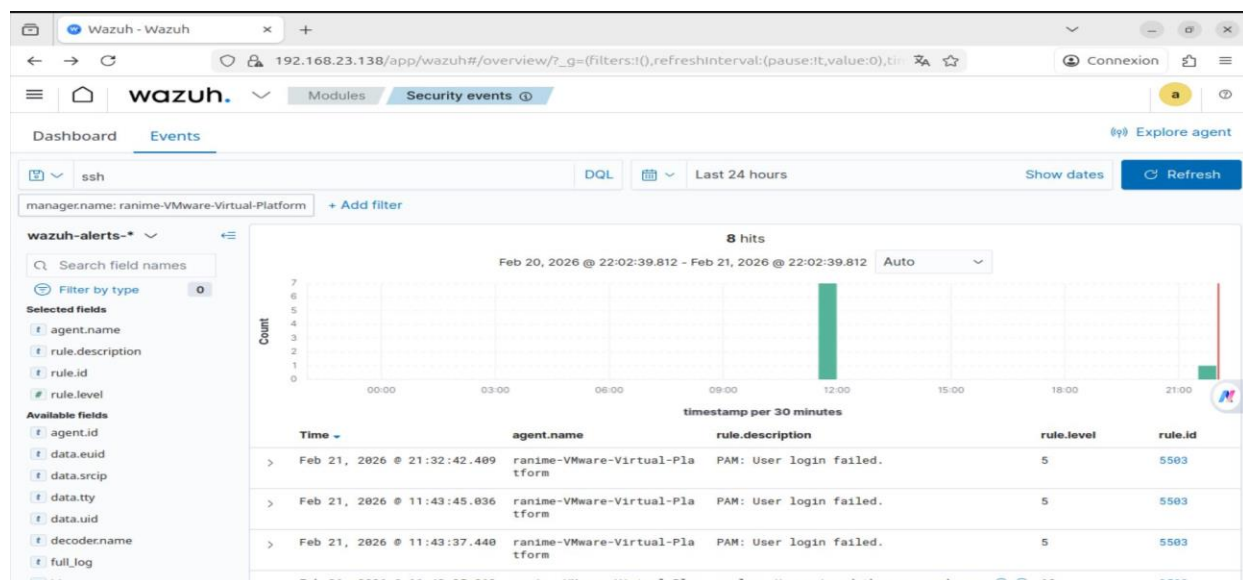


Figure 1: Wazuh Security Events Dashboard — SSH Alert Overview with MITRE ATT&CK Mapping

4.2 Agent Dashboard — kali-lab

The Kali Linux agent (ID 002) dashboard displayed aggregated compliance data (PCI DSS), MITRE top tactics, and a File Integrity Monitoring (FIM) events panel. The top MITRE tactic for this agent was Defense Evasion (count: 2).

PCI DSS compliance coverage was tracked across multiple requirements, with requirement 2.2 accounting for the largest share (49 events), followed by 10.6.1 (38 events).

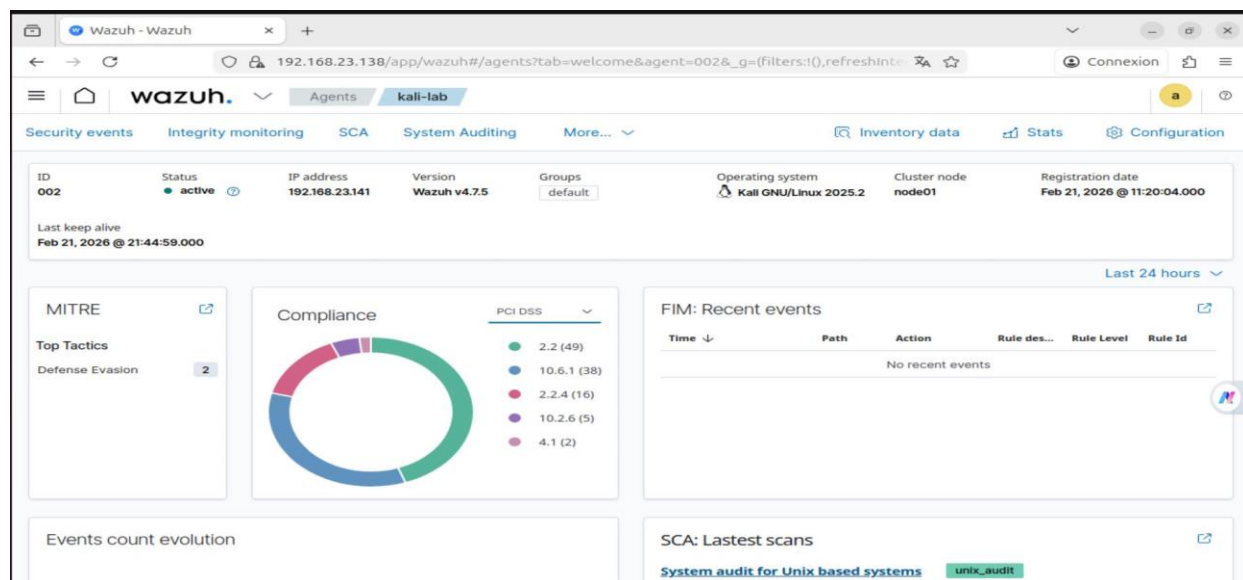


Figure 2: Wazuh Agent Dashboard — Kali Linux (kali-lab) — MITRE, Compliance & FIM Overview

4.3 Events Log — Detailed View

The Events tab with "ssh" filter displayed the 8 raw log entries, each timestamped and attributed to the agent. Key fields captured include agent name, rule description, rule level, and rule ID. A spike of 6 events was visible at 12:00 in the histogram, with a final alert at 21:00.

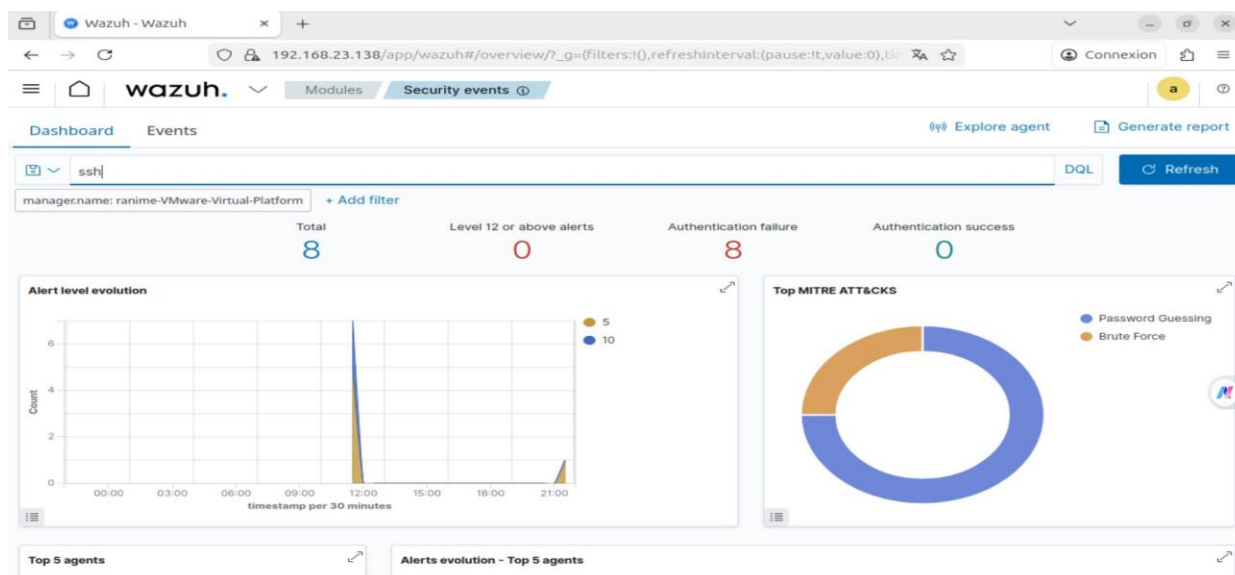


Figure 3: Wazuh Events Log — SSH Failed Login Attempts (8 hits, Feb 21 2026)

5. Key Findings & Analysis

5.1 Threat Detection

- Wazuh successfully detected all 8 SSH brute-force attempts in real time
- Alerts were correctly classified as authentication failures with rule 5503 (PAM: User login failed)
- The attack burst at 12:00 was clearly visible in the timeline histogram — demonstrating the value of time-series visualization for SOC analysts
- MITRE ATT&CK classification (Password Guessing + Brute Force) was automatically applied, enabling faster threat contextualization

5.2 Compliance Observations

- PCI DSS monitoring was active and logged 110 compliance-relevant events across the kali-lab agent
- No FIM (File Integrity Monitoring) changes were recorded during the session, confirming the attack remained confined to authentication attempts

5.3 Gaps Identified

- No Level 12+ (critical) alerts were generated — alert thresholds may need tuning for SSH brute force to escalate
- Active response (IP blocking) was not configured — a real SOC would automate this response
- The attacker machine (kali-lab) was monitored as a Wazuh agent, which is not representative of real-world black-box attacker scenarios

6. Skills Demonstrated

This lab provided hands-on practice in the following Tier-1 SOC analyst competencies:

- SIEM deployment and configuration (Wazuh)
- Agent onboarding and endpoint monitoring
- Real-time log collection and correlation
- Alert triage and severity classification
- Threat hunting using keyword and DQL filters
- MITRE ATT&CK framework mapping and analysis
- Dashboard interpretation and incident timeline reconstruction
- PCI DSS compliance monitoring

7. Recommendations

- Configure active response rules in Wazuh to auto-block IPs after N failed SSH attempts
- Raise alert rule levels for SSH brute force to ensure Level 12+ alerts trigger for critical events
- Add email or Slack notifications for high-severity alerts to simulate real SOC escalation workflows
- Expand the lab to simulate other attack vectors: web application attacks, privilege escalation, lateral movement
- Integrate Threat Intelligence feeds (VirusTotal, AbuseIPDB) into Wazuh for automated IoC enrichment

8. Conclusion

This SOC home lab successfully demonstrated end-to-end threat detection and monitoring using Wazuh SIEM. By simulating a real SSH brute-force attack from a Kali Linux machine against an Ubuntu victim, the lab provided practical experience in the core workflows of a Tier-1 SOC analyst.

The exercise validated that Wazuh can effectively detect, classify, and visualize malicious SSH activity in real time — mapping alerts to the MITRE ATT&CK framework automatically. The skills practiced here — log analysis, alert triage, SIEM navigation, and incident timeline reconstruction — are directly transferable to professional security operations roles.

— End of Report —